

CHAPTER 7: NETWORK SECURITY FUNDAMENTALS

COMMON NETWORK THREATS AND ATTACKS

1. Reconnaissance Attacks

A reconnaissance attack is not a direct theft; it is the process of gathering information. The attacker is trying to map out your network, find out which ports are open, and identify what operating systems you are running.

- **How it works:** Attackers use tools like **Nmap** or **Packet Sniffers** to send "pings" or packets to every possible IP address in your campus to see who responds.
- **Example:** An attacker outside the **Smart Campus** sends a "Port Scan" to your Campus-Edge router. They discover that the "Secure Faculty" zone has an open port for an old printer. Now they know exactly where to strike.

2. Denial of Service (DoS)

A DoS attack happens when an attacker sends so much "garbage" traffic to a specific device (like your Enrollment Server) that the device becomes overwhelmed and crashes. Legitimate users (your students) can no longer access it.

- **How it works:** One computer sends thousands of "Connection Requests" per second. The server tries to answer them all, runs out of memory (RAM), and stops working.

3. Distributed Denial of Service (DDoS)

A DDoS is a DoS attack on a massive scale. Instead of one computer attacking, the attacker uses a **Botnet**—thousands of infected computers (zombies) all over the world—to attack the target at the same time.

- **How it works:** The attacker sends a single command to their botnet, and suddenly, 10,000 computers start flooding the **Smart Campus** router with traffic.
- **Example:** During the final hour of student registration, a botnet targets the student management system's server. Because the server is being hit with millions of fake data requests, the real students can't receive their SMS notifications.

FIREWALLS AND ACCESS CONTROL LISTS (ACLs)

The Firewall (The Perimeter Security)

A Firewall is a dedicated security device (or software) that monitors and filters incoming and outgoing network traffic based on an organization's previously established security policies.

- **How it works:** It acts as a barrier between your trusted internal network (the Campus) and untrusted external networks (the Internet). It uses **Stateful Inspection**, meaning it doesn't just look at a packet; it remembers if you actually asked for that data.
- **Example:** If a hacker tries to send a file to your server from the outside, the Firewall sees that the server never requested that file and blocks it instantly.

Access Control Lists (ACLs)

ACLs are sets of rules applied to a **Router's interface**. They tell the router which packets to **Permit** and which to **Deny**. In Cisco networking, we categorize them into two main types:

A. Standard ACLs

Standard ACLs are the "simple" version. They only look at the **Source IP Address**. They don't care what kind of traffic it is (Web, Email, File Transfer); they only care *who* sent it.

- **Rule Range:** 1–99 and 1300–1999.
- **Placement:** Should be placed as close to the **Destination** as possible.
- **Example Command:** `access-list 10 deny 192.168.1.5` (Blocks all traffic from that specific computer).

B. Extended ACLs

Extended ACLs are much more precise. They check the **Source IP**, the **Destination IP**, the **Protocol** (TCP/UDP), and the **Port Number** (HTTP, FTP, etc.).

- **Rule Range:** 100–199 and 2000–2699.
- **Placement:** Should be placed as close to the **Source** as possible (to save bandwidth).
- **Example Command:** `access-list 101 permit tcp 192.168.1.0 0.0.0.255 any eq 80` (Allows a whole subnet to browse the web but nothing else).

SECURE DEVICE CONFIGURATION

1. Disabling Unused Services (Closing the Windows)

By default, network devices often have services running that you might not need (like HTTP servers, CDP, or small services like "echo").

- **The Security Risk:** Every running service is a potential **Attack Vector**. If a service has a bug or a default password, an attacker can use it to gain control of the device.
- **The "Harden" Strategy:** If you don't use it, **turn it off**. This reduces the "Attack Surface" of your Campus-Edge routers.

2. Telnet vs. SSH (The "Secret" Language)

When you or your students want to configure a router remotely (from the faculty office instead of standing in the server room with a console cable), you have two main choices. In a modern **Smart Campus**, only one is acceptable.

A. Telnet

Telnet was the original way to manage devices over a network.

- **How it works:** It sends everything—including your **username and password**—in **Plain Text**.

- **The Danger:** Anyone on the network using a "Packet Sniffer" (like Wireshark) can see exactly what you type.

B. SSH - Secure Shell

SSH is the modern replacement for Telnet.

- **How it works:** It creates an **encrypted tunnel** between your computer and the router.
- **The Security:** Even if someone "sniffs" the packets, they will only see scrambled gibberish. It uses mathematical algorithms to ensure only the router can "unlock" the message.

References:

- Cisco Networking Academy. Networking Concepts and Routing & Switching Modules. Cisco Systems, Inc.